

OPERATOR'S GUIDE



Document 770012

Revision 1

GE Signa® 1.5T 8-CHANNEL CARDIAC ARRAY

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USAI Part Number: 160188

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SECTION 1 – INTRODUCTION

1-1 Operator's Guide Information

The operator's guide provides information on how to use the 8-Channel Cardiac Array. The user should read this guide before using the coil for imaging. The user is advised to refer to the MR scanner manual in conjunction with this guide.

Direct any questions regarding this manual to:

USA Instruments, Inc.
1515 Danner Drive
Aurora, Ohio 44202

1-2 Use of Symbols

Coil users should refer to the operator's guide for caution and warning explanations, using the symbols defined below.



Cautions provide information on possible equipment hazards.



Warnings provide information on possible hazards to human beings.

1-3 Coil Classification

Class II equipment (the symbol is shown right).



Type BF applied part (the symbol is shown right).



Ordinary equipment.

Suitable for continuous operation.

1-4 Safety

1-4-1 Coil Operating Safety

Patient safety and comfort should be the primary concern during the scanning procedure. Always follow proper safety, operating and maintenance procedures to ensure that the patient is not exposed to electrical or mechanical hazards that may potentially cause injury.

1-4-2 Patient Safety



WARNING!

Prevent all cables from forming loops and coming into contact with the patient. Contact may result in burns to the patient.

Ensure that the patient is comfortably positioned.

Monitor the patient periodically. Stop the scan immediately if the patient reports any heating, burning or tingling sensations.

1-4-3 Equipment Safety

All personnel using the coil should be instructed in the proper connection and handling of the coil.

Personnel should observe all warnings and cautions that appear in this guide.



WARNING!

Auxiliary equipment (such as gating equipment, vital signs monitoring systems and RF coils) that have not been specifically tested and approved for use in the MR environment may result in burns or other injuries to the patient. This equipment may also interfere with the proper operation of the coil.



WARNING!

The coil is not suitable for use in the presence of flammable anesthetics in combination with air, oxygen or nitrous oxide.



CAUTION



Prevent cables from forming loops. Looping will degrade scan performance of the coil through RF coupling.

1-4-4 Electrical and Mechanical Safety



WARNING!

The coil consists of both electrical and mechanical components. Service personnel must have specialized training to ensure the safe operating condition of the coil. Therefore, only properly trained and qualified personnel should be authorized to service the coil.

The electrical and mechanical assemblies and parts of the coil must be used with care and should be routinely inspected.

Before using the coil, visually check it to ensure there is no external damage. Do not use the coil if the housing or cable is broken.

1-4-5 Modifications

Any changes or modifications to the coil must be approved and performed by USA Instruments prior to installation.

1-4-6 Reporting of Incidents

Users should contact USA Instruments immediately to report an incident and/or injury to a patient, operator or maintenance employee that occurred as a result of coil operation.

If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation is conducted.

1-5 Coil Description

The 8-Channel Cardiac Array is a high-resolution coil for the cardiovascular system. It works at 1.5T field strength and is capable of doing Array Spatial Sensitivity Encoding Technique (ASSET). This design reduces the breath-hold time, which will aide the patient to successfully hold their breath for the designated time while scanning. The coil is composed of two separate pieces: anterior and posterior. The posterior piece is rigid. The anterior piece is semi-flexible. Each section contains four elements, and the coil consists of 8 receive only channels.



CAUTION

United States federal law restricts this device to sale by, or on the order of, a physician.



Cables bearing this Coil ID label are compatible with software versions 10x and newer. With software versions 11x and newer, the system will generate an error when an incompatible coil is prescribed.

1-6 Coil Mode Configurations

The main goal of this document is to help the user optimize image quality using the 8-Channel Cardiac Array, focusing on the coil configuration, patient positioning and FOV selection. Improper use of the coil is the major cause of image artifacts.

8CH Cardiac: 8 coil elements are active for this mode, 4 at posterior and 4 at anterior (see *Figure 1*).

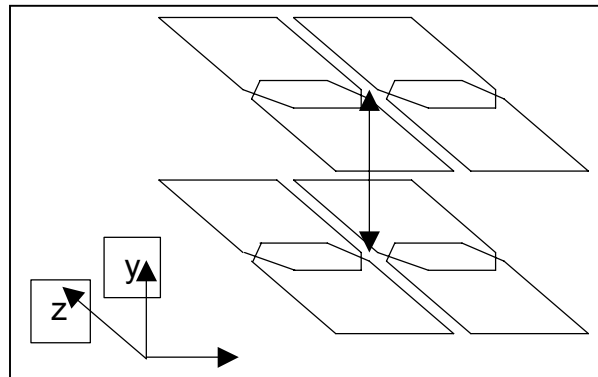


Figure 1: 8CH Cardiac active elements.

8CH Cardiac Anterior: 4 coil elements are active in the anterior section for this mode.

8CH Cardiac Posterior: 4 coil elements are active in the posterior section for this mode. The posterior section of the coil is detachable.

Element Dimensions

Each element is 14 cm x 14 cm.

The elements are separated in the R-L direction by 15 cm. The elements are critically overlapped in the S-I direction.

MAXIMUM RECOMMENDED FOV – TABLE 1-6

Modes	FOV
R-L Direction	30 cm
S-I Direction	30 cm
Diagonal Direction	42.5 cm

The 8-Channel Cardiac Array is optimized for ASSET applications in all directions, including obliques and double-obliques.

1-7 Environmental Requirements

ENVIRONMENTAL REQUIREMENTS – TABLE 1-7

	Transport and Storage	Operating Conditions
Barometric Pressure	645mm Hg to 795mm Hg	
Humidity	20% to 95% non-condensing	45% to 75% non-condensing
Temperature	-15° C to 55° C (5° F to 131° F)	15° C to 35° C (59° F to 95° F)

Dimensions

19.4" L x 16.5" W x 4.7" H (49.4cm x 42.0cm x 11.9cm)

Weight

12 lbs (5.4 kg)

1-8 Coil Specifications

COIL SPECIFICATIONS – TABLE 1-8

Field Strength	Coil Model	USAI Part Number	Frequency
1.5T	Spirit 9000	100391	63.87 MHz

1-9 Coil Installation and Service

1-9-1 Coil Installation Procedure

Coil installation requires the services of your GE Service Representative.

1-9-2 Coil Service

The coil cannot be serviced on site. Contact your authorized GE Service Representative for service and maintenance needs.

SECTION 2 – SETUP

2-1 Positioning the Coil



WARNING!

Remove any other coil or unused accessory device from the magnet before using the coil. Patient burns may result from coupling.

Step 1 – Position the posterior section on the cradle with the patient comfort pad (see *Figure 2*).

Step 2 – The coil connector and cable should always be routed directly to the magnet and should not form loops.

Step 3 - Place a clean cotton sheet over the coil so that the patient's skin will not come into direct contact with the coil.



Figure 2: Positioning the posterior section of the coil on the scanner cradle.

2-2 Positioning the Patient

Step 1 – Position the patient feet first, supine, with the region of interest lying on the posterior section of the coil. Position the anterior section above the region of interest. Reconnect the cable attaching the anterior and posterior sections of the coil, the cables should run between the patient's legs (see *Figure 3*).



Figure 3: Positioning the patient on the coil.

Step 2 – Position the anterior section above the anatomical region of interest.

Step 3 – Fasten the Velcro straps comfortably so that the coil is snug and comfortable.

Step 4 – Connect the anterior coil connector into the coil receptacle.

Step 5 – Connect the posterior coil connector into the anterior coil connector.

2-3 Localizing Anatomy

Position the alignment light on the region of interest and move the patient to the center of the magnet. The alignment lines on the coil indicate the center of the coil.

2-4 Choosing the Correct Coil Mode Configuration

Refer to Section 1-6 for detailed descriptions of each coil mode. Choose the coil mode most appropriate for the corresponding anatomy of interest.

The 8-Channel Cardiac Array is provided with external marks that represent each coil element.

Step 1 – Landmark on the coil marker, not on the anatomy.

Step 2 – Select the appropriate number of elements according to the area that needs to be covered.

Step 3 – Landmark on an imaginary middle line, between the elements used.

Step 4 – Considering that each element is 12cm, select a coil configuration and FOV based on the number of elements chosen.

To reduce artifacts in sagittal or coronal imaging, use the minimum number of coil sections to image the given FOV. For any 4-section coil selection, the maximum FOV is 48cm. For any 3-section selection, the maximum FOV is 36cm. For any 2-section selection, the maximum FOV is 25cm.

2-5 Setting up the Scan

Note: Auto-N-Coil is only compatible with 4-Channel Cardiac coils and will disable elements in the middle of a given series producing undesired image artifacts.

The coil is designed to be compatible with recommended protocols suggested by the system's manufacturer. Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

The procedure described in section 2-4 is seen as the best practice for the coil usage. If you are still unsure of coil configuration selection for the region of interest the control variable APA will determine the best choice for FSE sequences only.

Step 1 – Start with a coil selection of either: **8CARDIAC**, **8CARDIACANT** or **8CARDIACPOST**.

Step 2 – Enable the APA feature under the User CV page.

Step 3 – Based on the landmark and FOV, the scanner will determine which coils should be used for optimal image quality.

The image annotation does not record the selected coils by APA.

2-6 Recommended Scan Protocols

The coil is designed to be compatible with recommended protocols suggested by the system's manufacturer.

Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	USAI Part #	Qty
Phantom Kit	2304255-3	155158	1

3-2-2 Explanation of Procedure

The image quality check uses two consecutive acquisitions for the same slice. The signal is measured at the center of the phantom with a ROI of about 60% of the phantom size. The noise is measured on the subtracted image of the two consecutive signal images using the same ROI. SNR is calculated by dividing the Signal mean by the noise standard deviation. All measurements are performed on composite images. Interim images are acquired for reference only.

3-2-3 SNR Setup and Test Execution for 11.0 M3A and Greater Software Versions

1. Remove any other surface coils (if present) from the cradle.
2. Place the coil on the patient table. Position the phantom set on the coil without the pad, lining up the pegs on the phantom to the holes in the posterior section of the coil (see *Figure 4*).



Figure 4: Positioning the phantom on the coil.

3. Using the raised notches on top of the phantom, align the anterior section on the phantom and fasten the Velcro straps.

4. Landmark the coil at the indicated line and advance to scan (see *Figure 5*).



Figure 5: Landmarking the coil.

5. From the service browser, select **Image Quality** and **Coil SNR**, then **Click here to start this tool.**
6. Select the coil type you want to run, enter the serial number and select **Start Scan**.
Note: Additional documentation can be accessed by selecting **File, Help, [coil type]**.

SNR Data Sheet

Use the table provided below to record the pass/fail results of the SNR test obtained from the Functional Checks section.

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

SECTION 4 – MAINTENANCE

4-1 Coil Care



WARNING!

Detach coil connector from the scanner before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning until the coil has dried completely. Electric shock may result if the coil is attached to the system during cleaning or when it is wet.



CAUTION

Do not spray or pour the cleaning solution directly on the coil. Do not submerge the coil in the solution. The coil contains sensitive electronic components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: (1) a ten percent bleach solution (some discoloration may occur), (2) one ounce commercial dishwashing liquid mixed with one gallon of water or (3) warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning the coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

4-3 Replaceable Accessories

The table below lists coil accessories that may be purchased for replacement.

REPLACEABLE ACCESSORIES LIST – TABLE 4-3

Part Name	GE Part #	USAI Part #
Coil	2304255-2	165148
Anterior Cable Assembly	2304255-7	155156
Posterior Cable Assembly		155157
Phantom Kit	2304255-3	155158
QA Phantom		155159
Upper Annular Ring		155160
Lower Annular Ring		155161
Upper/Lower Phantom Positioner		155162
Middle Phantom Positioner		155163
Patient Comfort Pad	2304255-6	155164
Patient Headrest Pad		155165
Patient Lower Body Pad		155166

For additional information, contact:

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REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
1	10/03	Teresa DeMarco	First Issue